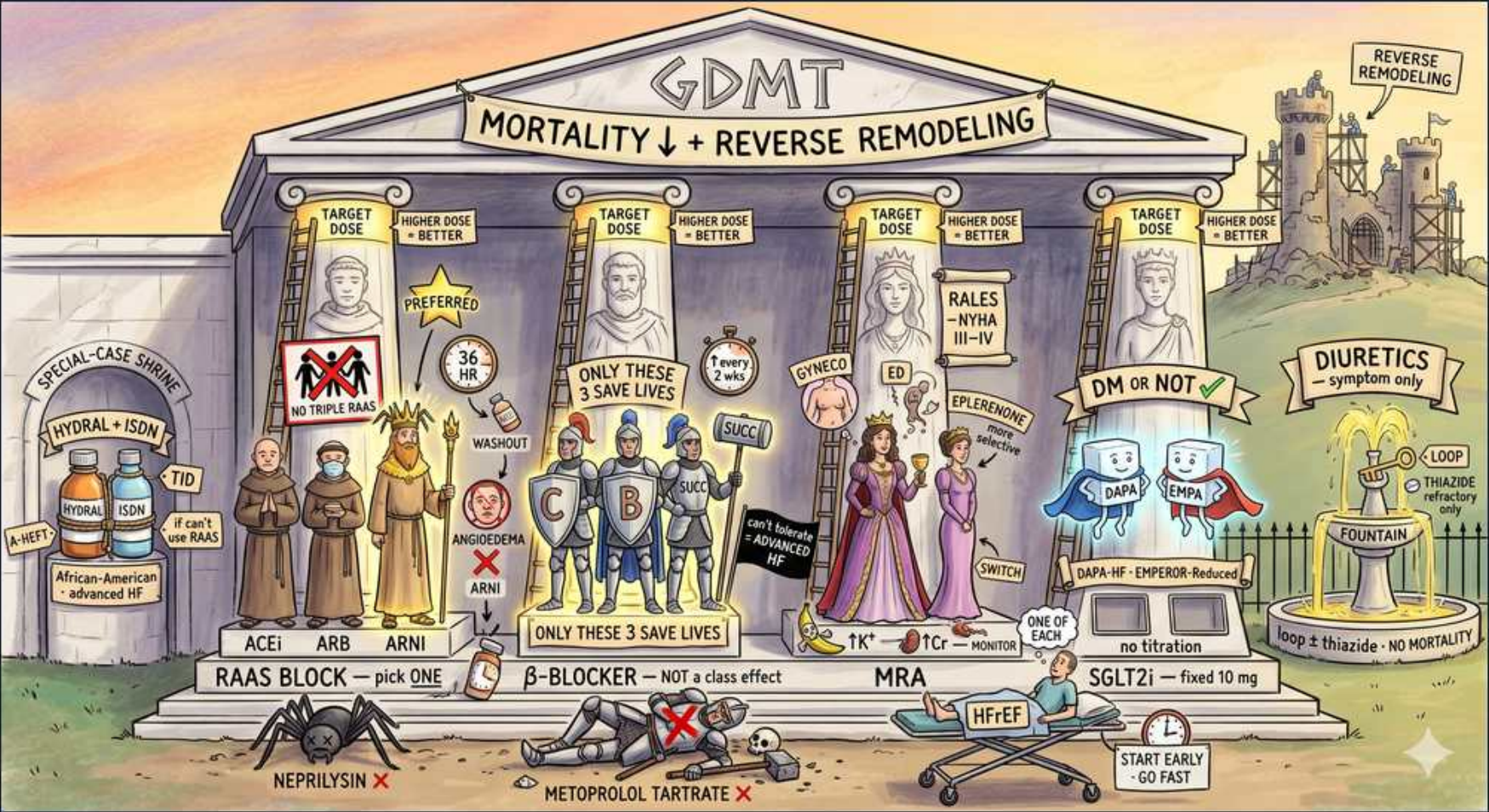


Heart Failure — SCENE 4 — HFrEF GDMT: The Four-Pillar Apothecary



1. GDMT goal: mortality ↓ + reverse remodeling

WHAT YOU SEE

Temple banner 'MORTALITY ↓ + REVERSE REMODELING'

WHAT IT ENCODES

The four foundational pillars of HFrEF (LVEF ≤40%) GDMT — ARNI/ACEi/ARB, evidence-based beta-blocker, MRA, and SGLT2i — each independently reduce all-cause mortality and HF hospitalization, and together produce reverse remodeling (LVEF improvement, reduced LV volumes). Combined use confers roughly a 60–70% relative mortality reduction vs none. Modern teaching: start all four EARLY and rapidly (within weeks, even in-hospital) rather than the old slow sequential approach.

BOARD TRAP

Trap: a stem implies that improving symptoms (with diuretics) equals improving survival. WRONG — only the four pillars (plus, in selected groups, hydralazine/ISDN and device therapy) reduce mortality; diuretics, digoxin, and ivabradine are symptom/hospitalization tools without a mortality benefit. Discriminator: 'reduces mortality / reverse remodeling' should map you to the four pillars, never to a diuretic.

2. Pillar 1: RAAS block — ARNI > ACEi > ARB

WHAT YOU SEE

First pillar 'RAAS BLOCK — pick ONE: ACEi / ARB / ARNI'

WHAT IT ENCODES

Pillar 1 = RAAS inhibition; choose ONE agent. Preference order: ARNI (sacubitril/valsartan) > ACEi > ARB. ARNI (PARADIGM-HF) was superior to enalapril for CV death/HF hospitalization. Use an ACEi (e.g., lisinopril, enalapril) if ARNI is unavailable; reserve an ARB (valsartan, candesartan, losartan) for ACEi-intolerant (cough/angioedema) patients. Start low and titrate every 1–2 weeks to target; monitor K+, creatinine, and BP.

BOARD TRAP

Trap: combining ACEi + ARB (or any two RAAS agents) for 'additive' benefit. WRONG — dual RAAS blockade markedly increases hyperkalemia, hypotension, and acute kidney injury without survival benefit (ONTARGET). Discriminator: pick exactly ONE RAAS agent; if switching ACEi→ARNI, do not overlap (see washout).

3. ARNI preferred / 36h washout

WHAT YOU SEE

'PREFERRED', '36 hr WASHOUT' note on the ARNI pillar

WHAT IT ENCODES

ARNI = sacubitril (neprilysin inhibitor) + valsartan; start 49/51 mg BID (or 24/26 mg BID if low BP, elderly, renal/hepatic impairment, or ACEi/ARB-naïve) and double toward target 97/103 mg BID. When switching from an ACEi, observe a 36-HOUR WASHOUT before the first ARNI dose because overlapping ACE + neprilysin inhibition stacks bradykinin and causes angioedema. No washout is needed switching from an ARB.

BOARD TRAP

Trap: starting ARNI the morning after the last lisinopril dose (or co-administering them). WRONG — you must wait 36 hours after the last ACEi dose; failing this raises angioedema risk. Discriminator: ACEi→ARNI = 36h washout; ARB→ARNI = no washout. Also avoid ARNI within 36h of an ACEi for the same reason.

4. No triple RAAS / angioedema warning

WHAT YOU SEE

Red 'X NO TRIPLE RAAS' and 'ANGIOEDEMA' caution

WHAT IT ENCODES

Never combine ACEi + ARB + ARNI (these all block the RAAS axis): triple/dual blockade causes hyperkalemia, hypotension, and renal failure with no added benefit. ARNI and ACEi are absolutely contraindicated with any prior history of angioedema; ARNI is also contraindicated in pregnancy and with concomitant aliskiren in diabetics. Hold/avoid in bilateral renal artery stenosis.

BOARD TRAP

Trap: prescribing an ARNI to a patient who had ACEi-associated angioedema, assuming the new drug class is safe. WRONG — prior angioedema (to ACEi or otherwise) contraindicates ARNI because neprilysin inhibition independently raises bradykinin. Discriminator: history of angioedema → choose an ARB-based regimen or another pillar, not ARNI or ACEi.

5. Pillar 2: Beta-blocker (only 3 save lives)

WHAT YOU SEE

Three knights 'C B B' = Carvedilol, Bisoprolol, metoprolol succinate

WHAT IT ENCODES

Pillar 2 = a mortality-proven beta-blocker — ONLY carvedilol (target 25 mg BID, 50 mg BID if >85 kg), bisoprolol (target 10 mg daily), and metoprolol SUCCINATE (extended-release, target 200 mg daily). They blunt chronic sympathetic toxicity and promote reverse remodeling (CIBIS-II, MERIT-HF, COPERNICUS). Start LOW only when EUVOLEMIC/stable and 'start low, go slow,' doubling every ~2 weeks; transient symptom worsening can occur early.

BOARD TRAP

Trap: 'all beta-blockers are equivalent in HFrEF (class effect).'

WRONG — it is NOT a class effect; atenolol, metoprolol TARTRATE, propranolol, and nebivolol (in the US) are not first-line mortality agents for HFrEF. Discriminator: only carvedilol, bisoprolol, and metoprolol succinate. Also never START or uptitrate a beta-blocker during acute decompensation/cardiogenic shock.

6. Metoprolol tartrate ☐

WHAT YOU SEE

Fallen knight 'METOPROLOL TARTRATE ☐' at base

WHAT IT ENCODES

Metoprolol TARTRATE (immediate-release, dosed BID) is NOT a proven mortality-reducing beta-blocker for HFrEF — the MERIT-HF survival data are specific to metoprolol SUCCINATE (CR/XL, once daily). If a patient is on tartrate, convert to succinate for the chronic HFrEF mortality benefit (tartrate is acceptable for acute rate control/peri-MI situations).

BOARD TRAP

Trap: a HFrEF patient on metoprolol tartrate is described as 'already on guideline beta-blocker therapy,' so the answer is no change. WRONG — switch tartrate to succinate (or to carvedilol/bisoprolol) to obtain the mortality benefit. Discriminator: succinate = once-daily, mortality-proven; tartrate = twice-daily, not proven for chronic HFrEF survival.

7. Pillar 3: MRA (NYHA II-IV)

WHAT YOU SEE

Queen 'MRA' with 'RALES — NYHA III-IV', spironolactone/eplerenone

WHAT IT ENCODES

Pillar 3 = a mineralocorticoid receptor antagonist (spironolactone or eplerenone), which reduces mortality and HF hospitalization in HFrEF (RALES in NYHA III–IV; EMPHASIS-HF starting at NYHA II) and post-MI LV dysfunction (EPHESUS). Typical: spironolactone start 12.5–25 mg daily → target 25–50 mg; eplerenone 25 → 50 mg daily. Eligibility gate: K+ <5.0 mEq/L and eGFR >30 mL/min; recheck K+/creatinine at ~3 days, 1 week, then periodically.

BOARD TRAP

Trap: starting/continuing an MRA when K+ is 5.4 or eGFR is 25, or in a patient on potassium supplements/high-K diet. WRONG — this courts life-threatening hyperkalemia (the major RALES real-world harm). Discriminator: MRA requires K+ <5.0 and eGFR >30; hold for K+ ≥5.5 and recheck labs. Don't combine an MRA with another potassium-sparing agent or NSAID.

8. MRA: monitor K+ / gynecomastia

WHAT YOU SEE

'↑K+' warning, 'GYNECO' (spironolactone), eplerenone alternative

WHAT IT ENCODES

MRA key adverse effects: HYPERKALEMIA and worsening renal function (monitor K+/Cr), plus spironolactone's antiandrogenic effects — painful GYNECOMASTIA, breast tenderness, and menstrual irregularity (up to ~10% in RALES). Eplerenone is the selective MRA and largely avoids these endocrine effects, so switch a man who develops gynecomastia from spironolactone to eplerenone.

BOARD TRAP

Trap: a man on spironolactone develops gynecomastia and the answer is to stop all MRA therapy (losing mortality benefit). WRONG — switch to eplerenone, which preserves the MRA mortality benefit without the antiandrogenic side effects. Discriminator: gynecomastia is a spironolactone-specific (off-target) effect, not a reason to abandon the MRA pillar.

9. Pillar 4: SGLT2i (DM or not)

WHAT YOU SEE

Blue bottles DAPA/EMPA, 'DM or NOT?'

WHAT IT ENCODES

Pillar 4 = an SGLT2 inhibitor — dapagliflozin (DAPA-HF) or empagliflozin (EMPEROR-Reduced) — which reduces CV death and HF hospitalization in HFrEF REGARDLESS of diabetes status. Benefit is rapid (within weeks) and independent of glycemic control, driven by natriuresis, reduced preload/afterload, and cardiac metabolic effects. Watch for genital mycotic infections, volume depletion, and (rarely) euglycemic DKA — hold during acute illness/surgery (sick-day rules).

BOARD TRAP

Trap: 'SGLT2i only help HFrEF patients who have diabetes.' WRONG — the HF mortality/hospitalization benefit is INDEPENDENT of diabetes (non-diabetic patients benefited equally in DAPA-HF and EMPEROR-Reduced). Discriminator: prescribe an SGLT2i in any eligible HFrEF patient (eGFR permitting, generally ≥20–25), diabetic or not.

10. SGLT2i: fixed 10mg, no titration

WHAT YOU SEE

'no titration', 'fixed 10 mg' label

WHAT IT ENCODES

SGLT2i are given at a FIXED dose with NO up-titration: dapagliflozin 10 mg daily or empagliflozin 10 mg daily — uniquely simple among the pillars, which makes them easy to start early/in-hospital. There is a small expected reversible early eGFR dip that should not prompt discontinuation.

BOARD TRAP

Trap: a stem reports a small early creatinine bump after starting an SGLT2i and the answer is to stop the drug. WRONG — the initial eGFR dip is hemodynamic and reversible; continue the drug for its cardiorenal benefit. Discriminator: unlike RAAS/BB, SGLT2i need no titration and the early eGFR dip is expected, not a stop signal (hold only for acute illness/DKA risk).

11. Diuretics = symptom only, no mortality

WHAT YOU SEE

Right gazebo 'DIURETICS — symptom only', fountain

WHAT IT ENCODES

Loop diuretics (furosemide, torsemide, bumetanide) relieve congestion/volume overload and improve symptoms but do NOT reduce mortality; use the LOWEST effective dose to maintain euolemia. Torsemide/bumetanide have more reliable oral absorption than furosemide in gut edema. Add a thiazide (metolazone) for diuretic resistance (sequential nephron blockade), monitoring for hypokalemia/hyponatremia and azotemia.

BOARD TRAP

Trap: 'loop diuretics improve survival in heart failure.' WRONG — diuretics treat congestion/symptoms only and have no proven mortality benefit; over-diuresis can worsen renal function and activate the RAAS. Discriminator: if a question asks which therapy reduces mortality, never pick a diuretic — pick one of the four pillars (or hydralazine/ISDN in Black patients).

12. Hydralazine + ISDN (Black patients)

WHAT YOU SEE

Special-case shelf 'HYDRAL + ISDN, African-American, advanced HF'

WHAT IT ENCODES

Hydralazine + isosorbide dinitrate (balanced arterial + venous vasodilation; nitrate supplies NO, hydralazine limits nitrate tolerance) adds a mortality benefit ON TOP of standard GDMT in self-identified Black/African-American patients with NYHA III–IV HFrEF (A-HeFT). It is also the go-to RAAS alternative when ACEi/ARB/ARNI are contraindicated (e.g., advanced CKD, hyperkalemia, prior angioedema, pregnancy).

BOARD TRAP

Trap: choosing hydralazine/ISDN as a first-line substitute for RAAS blockade in the general HFrEF population. WRONG — its proven additive mortality benefit is in Black patients with advanced (NYHA III–IV) HFrEF already on GDMT; in others it's mainly a fallback when RAAS inhibitors can't be used. Discriminator: the trial-supported benefit is population-specific (A-HeFT).

13. Titrate to target dose

WHAT YOU SEE

Repeated 'TARGET DOSE / HIGHER DOSE = BETTER' columns on every pillar

WHAT IT ENCODES

For RAAS inhibitors, beta-blockers, and MRAs, uptitrate to the maximally tolerated/target dose used in the trials, because higher doses yield greater benefit; reassess and double roughly every 1–2 weeks (beta-blockers when stable/euvolemic). SGLT2i are the exception — fixed dose, no titration. Monitor BP, HR, K⁺, and renal function at each step.

BOARD TRAP

Trap: leaving a patient indefinitely on a 'starter' low dose because they are tolerating it without symptoms. WRONG — failing to uptitrate to target deprives the patient of full mortality benefit (a common real-world quality gap). Discriminator: tolerance without side effects is a reason to KEEP titrating toward target, not to stop.

14. Start early, go fast

WHAT YOU SEE

Stretchers 'HFrEF', 'START EARLY — GO FAST'

WHAT IT ENCODES

Modern strategy (e.g., STRONG-HF): initiate ALL FOUR pillars early and rapidly — often simultaneously and even before/at hospital discharge — then up-titrate quickly with close follow-up (labs and a visit within ~1–2 weeks). Rapid quadruple therapy reduced death/readmission vs. slow usual care. Order can be individualized (e.g., low-BP patients may lead with SGLT2i + a low-dose pillar).

BOARD TRAP

Trap: the old sequential dogma — fully maximize one agent before adding the next, taking months. WRONG by current guidelines — early simultaneous initiation of all four pillars saves more lives than slow sequential build-up. Discriminator: 'start low/go slow' applies to titration of an individual beta-blocker, NOT to delaying the other three pillars.
